CLINICAL EXERCISE PRESCRIPTION LIBRARY

Hydrotherapy Protocols (Medium & High Resistance)

Aquatic Buoyancy (30°C–34°C), Hydrostatic Unloading & Band Resistance Protocols

---

Clinical Governance Team:

• Lead Author: Physiotherapist Specialist (PHYSIO)
• Surgical & Biomechanical Oversight: Surgeon Persona (SURGEON)
• Neuro-Psychiatric Review: Neuro-PPS Persona (NEURO_PPS)
• Evidence Synthesis: Medical Researcher Persona (RESEARCHER)
• Clinical Coordination: Medical Overseer (MDT_LEAD)

Prescription Date: 2026-10-12 | **Document Version:** 1.0.0

How to Use This Booklet:

Perform in warm water (30°C–34°C) at chest depth to achieve 60–75% joint axial decompression. Always anchor bands to solid pool rails. Never elevate arms above 80°.

Included Clinical Protocols:

[H-MED1] [H-MED2] [H-EXP1] [H-EXP2] [H-EXP3]

MANDATORY CLINICAL RESTRICTIONS:

• **Cervical Spine (C4–C7 Fusion)**: Strict neutral alignment. Zero forced end-range extension or neck rotation. No classical Pilates head curls (head grounded).
• **Right Shoulder (Subacromial/C7–T1)**: Active elevation strictly limited below 80°–90°. Elbows tucked below shoulder line during all rows and presses.
• **Lower Chain (Left Knee Osteotomy & Right Ankle)**: Braced on land. Max 45° shallow knee bend. Zero crouching or kneeling directly on unpadded surfaces.

TABLE OF CONTENTS & PROTOCOL OVERVIEW

Review protocol parameters, equipment setup, and durations before initiating sessions.

ID	PROTOCOL TITLE	DURATION	INTENSITY	EQUIPMENT REQUIRED
H-MED 1	30-Minute Dynamic Aqua Balance & Banded Core	30 Minutes	**Medium**	Heavy/Medium Theraband, Pool Rail, Flotation Noodle
H-MED 2	45-Minute Total Body Aquatic Conditioning	45 Minutes	**Medium**	Therabands, Pool Rail, Kickboard
H-EXP1	60-Minute Heavy Hydro Resistance & Power Protocol	60 Minutes	**Expert / High Resistance**	Heavy Therabands (Black/Silver), Pool Rail, Hydro Dumbbells, Flotation Belt
H-EXP2	75-Minute Multi-Planar Hydro Endurance & Kinetic Chain Overhaul	75 Minutes	**Expert / High Resistance**	Heavy Loop Bands, Pool Rail, Water Dumbbells, Pool Noodle
H-EXP3	90-Minute Master Hydro Athlete Rehabilitation Protocol	90 Minutes	**Expert / Maximum Resistance**	Full Hydro Resistance Kit (Heavy bands, Hydro paddles, Flotation belt, Noodles, Rail)

MANDATORY CLINICAL RESTRICTIONS:

• **Cervical Spine (C4–C7 Fusion)**: Strict neutral alignment. Zero forced end-range extension or neck rotation. No classical Pilates head curls (head grounded).
• **Right Shoulder (Subacromial/C7–T1)**: Active elevation strictly limited below 80°–90°. Elbows tucked below shoulder line during all rows and presses.
• **Lower Chain (Left Knee Osteotomy & Right Ankle)**: Braced on land. Max 45° shallow knee bend. Zero crouching or kneeling directly on unpadded surfaces.

H-MED1: 30-Minute Dynamic Aqua Balance & Banded Core

[MEDIUM]
Duration: 30 Minutes

Equipment Needed: Heavy/Medium Theraband | Pool Rail | Flotation Noodle

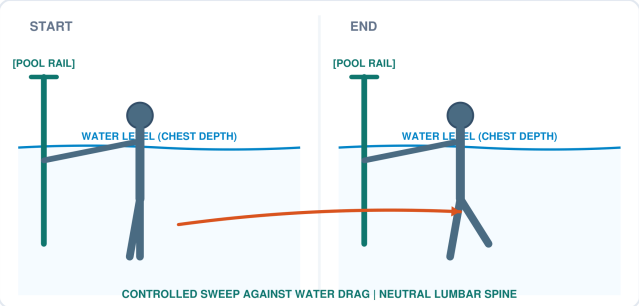
S1: 5m	S2: 8m	S3: 7m	S4: 6m	S5: 4m

Step 1: Warm-Up

Dosage: 5 mins

Instructional Cues:
- Heel-to-toe stride
- Hold pool rail for balance
- Chest-depth water support

Source: Forward/backward water walking (heel-to-toe), side-stepping holding pool rail.



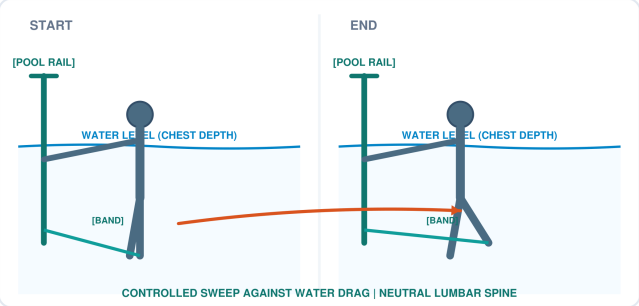
SAFETY MANDATE: Keep spine upright; avoid twisting at knees or ankles.

Step 2: Submerged Banded Hip Abduction & Extension

Dosage: 8 mins (3 sets x 12 reps per leg)

Instructional Cues:
- Slow 2-second outward sweep
- Resist water drag
- Keep torso upright and stable

Source: Band looped around pool rail and ankle. 3 sets x 12 reps per leg, slow 2-second outward sweep against water drag.



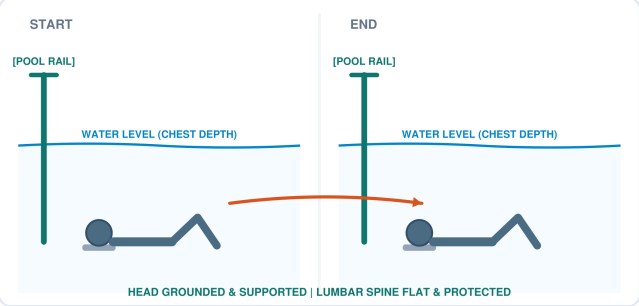
SAFETY MANDATE: Loop band securely to solid pool rail only; do not hyperextend lumbar spine.

Step 3: Rail-Supported Hydro Pelvic Tilts & Core Neutral Holds

Dosage: 7 mins (Hold 10s x 6 reps)

Instructional Cues:
- Back flat against pool wall
- Engage deep transverse core
- Hold 10 seconds per rep

Source: Back to pool wall, engage deep core and flatten lumbar curve into wall. Hold 10s x 6 reps.



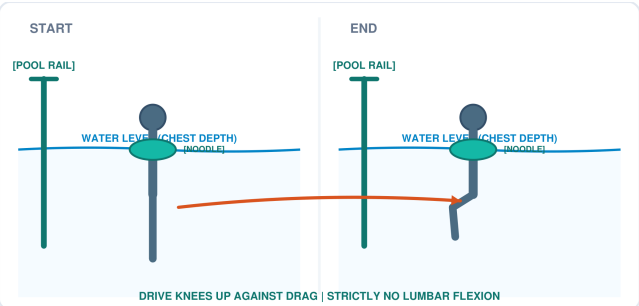
SAFETY MANDATE: Preserve neutral cervical posture against pool wall; no straining.

Step 4: Noodle Supported Knee Drives

Dosage: 6 mins (3 sets x 15 reps)

Instructional Cues:
- Noodle under armpits
- Drive knees up smoothly
- Engage lower abdominals

Source: Float with noodle under axillae, drive knees toward chest against water resistance (no lumbar flexion). 3 sets x 15 reps.



SAFETY MANDATE: Strictly no lumbar flexion during knee drive; maintain neutral spinal posture.

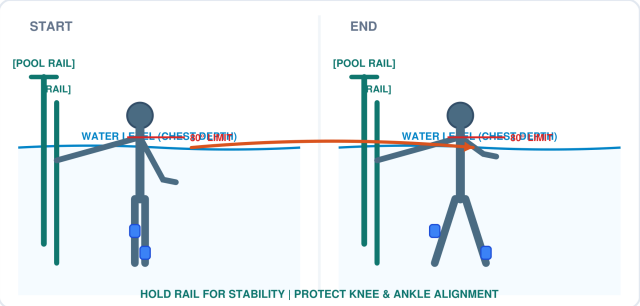
Step 5: Cool-Down

Dosage: 4 mins

Instructional Cues:

- Gentle water walking
- Deep diaphragmatic breathing
- Downregulate heart rate

Source: Water walking with deep diaphragmatic breathing.



SAFETY MANDATE: Maintain calm, rhythmic breathing without breath holding.

H-MED2: 45-Minute Total Body Aquatic Conditioning

[MEDIUM]
Duration: 45 Minutes

Equipment Needed: Therabands | Pool Rail | Kickboard

S1: 10m	S2: 10m	S3: 10m	S4: 10m	S5: 5m

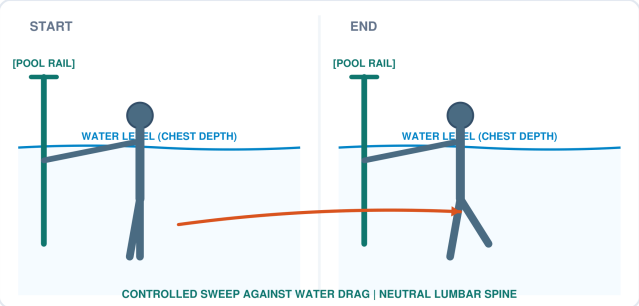
Step 1: Aquatic Gait & Scapular Activation

Dosage: 10 mins

Instructional Cues:

- Submerged arm sweeps
- Isometric chin retractions
- Forward/backward gait

Source: Forward/backward gait drills, submerged water sweeps and chin retractions.



SAFETY MANDATE: Cervical spine strictly in neutral; chin tucks in small pain-free range.

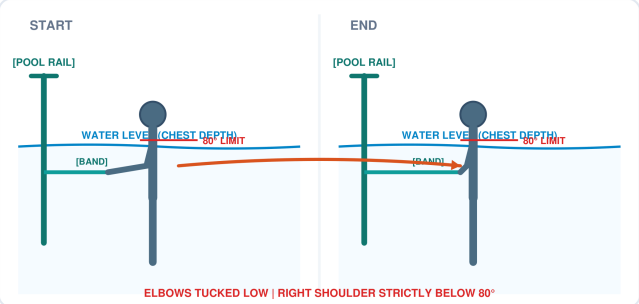
Step 2: Submerged Banded Low Rows & W-Rows

Dosage: 10 mins (3 sets x 12 reps)

Instructional Cues:

- Elbows tucked low
- Squeeze scapulae back into water
- Waist-height band anchor

Source: Band anchored to pool rail at waist level. Elbows tucked low, squeeze scapulae back into water. 3 sets x 12 reps.



SAFETY MANDATE: Keep elbows below shoulder line; do not elevate shoulders or trigger C7-T1 burning.

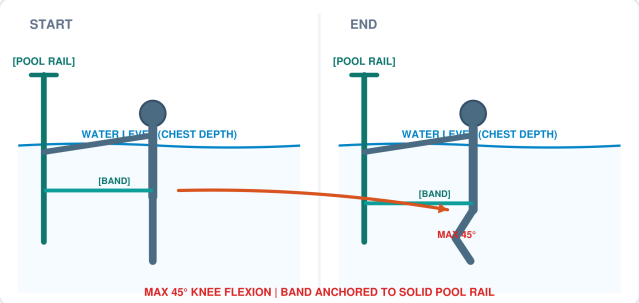
Step 3: Rail-Supported Banded Shallow Squat Presses

Dosage: 10 mins

Instructional Cues:

- Max 45° knee flexion
- Drive through glutes
- Band around waist anchored to rail

Source: Band around waist anchored to rail. Shallow supported mini-squats (max 45° knee flexion), driving through glutes.



SAFETY MANDATE: Strict limit of 45° knee bend; no deep crouching or knee shearing.

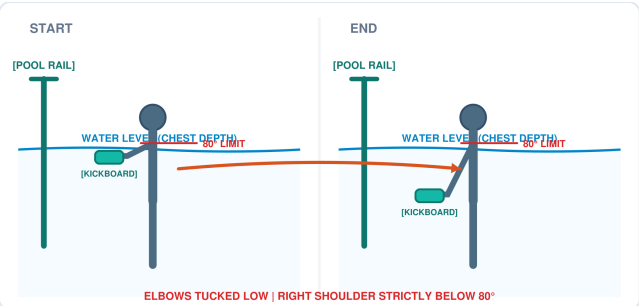
Step 4: Kickboard Aquatic Press-Downs

Dosage: 10 mins (3 sets x 10 reps)

Instructional Cues:

- Submerge kickboard at chest level
- Press downward engaging lats
- Brace anterior core

Source: Submerge kickboard with both hands at chest level, pressing downward to engage lats and anterior core. 3 sets x 10 reps.



SAFETY MANDATE: Keep wrists neutral; maintain shoulder elevation strictly below 80°.

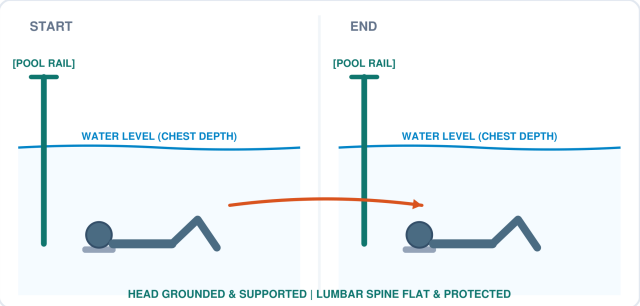
Step 5: Buoyancy Decompression

Dosage: 5 mins

Instructional Cues:

- Neck roll supporting cervical curve
- Bolster under knees
- Full buoyant floating

Source: Supported supine float with neck roll and knee bolster.



SAFETY MANDATE: Ensure full cervical support; neck completely relaxed in neutral.

H-EXP1: 60-Minute Heavy Hydro Resistance & Power Protocol

[EXPERT / HIGH RESISTANCE]

Duration: 60 Minutes

Equipment Needed: Heavy Therabands (Black/Silver) | Pool Rail | Hydro Dumbbells | Flotation Belt

S1: 10m	S2: 10m	S3: 10m	S4: 12m	S5: 10m	S6: 8m

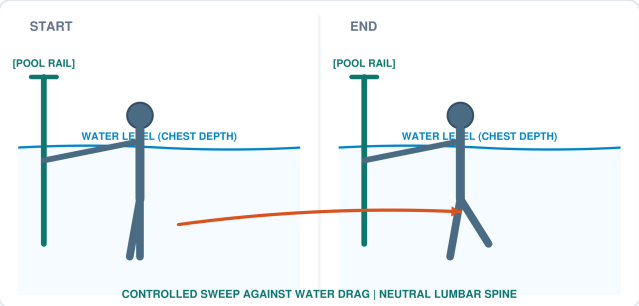
Step 1: Dynamic Thermal & Gait Warm-Up

Dosage: 10 mins

Instructional Cues:

- High-knee water marching
- Multi-planar lateral steps
- Submerged pendulum sweeps

Source: High-knee water marching, multi-planar lateral walk-outs, gentle submerged arm pendulum sweeps.



SAFETY MANDATE: Arm sweeps kept below 80° shoulder elevation; controlled foot placement.

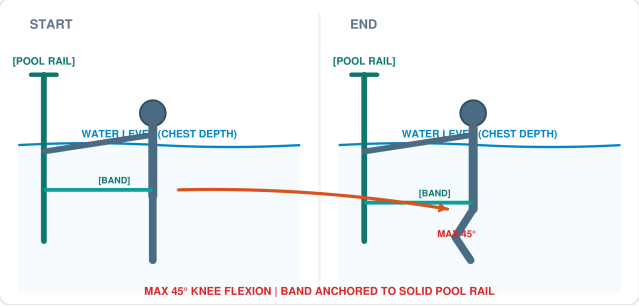
Step 2: Banded Pool-Rail Power Squat Pulses

Dosage: 10 mins (4 sets x 12 reps)

Instructional Cues:

- Heavy band at pelvis
- Controlled shallow pulses
- Push against band resistance

Source: Heavy band anchored to pool rail and pelvis. Controlled shallow squat pulses against band tension and water buoyancy. 4 sets x 12 reps.



SAFETY MANDATE: Shallow pulses only (max 45° knee flexion); protect left osteotomy site.

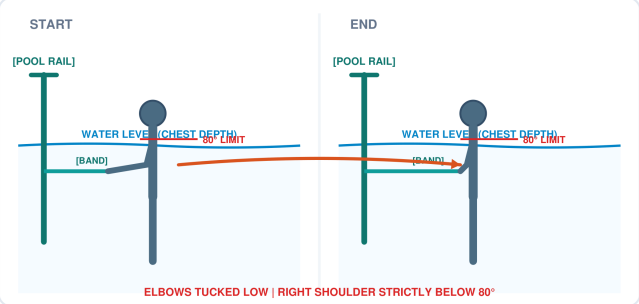
Step 3: Submerged Unilateral Band Rows & Chest Press

Dosage: 10 mins (3 sets x 12 reps)

Instructional Cues:

- Left arm powerful rows
- Right arm micro-pulls below 80°
- Torso square to rail

Source: Left arm heavy explosive rows; right arm controlled micro-range pulls to tolerance below 80°. 3 sets x 12 reps.



SAFETY MANDATE: Right shoulder limited below 80°; stop immediately if C7-T1 burning occurs.

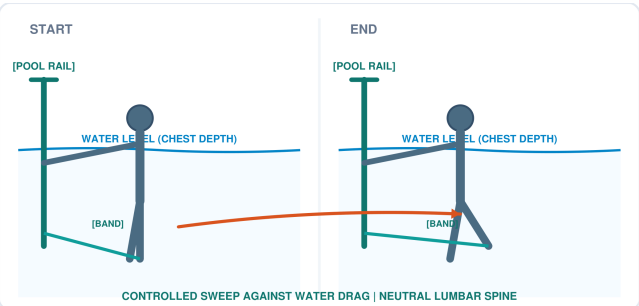
Step 4: Banded Submerged Leg Extensions & Lateral Kicks

Dosage: 12 mins (3 sets x 12 reps per side)

Instructional Cues:

- Heavy band on ankle
- Full quad & glute contraction
- Resist water turbulence

Source: Heavy ankle band anchored to rail. Resist water turbulence through full quad and glute contraction. 3 sets x 12 reps per side.



SAFETY MANDATE: Anchor securely to solid pool rail; avoid knee hyperextension or twisting.

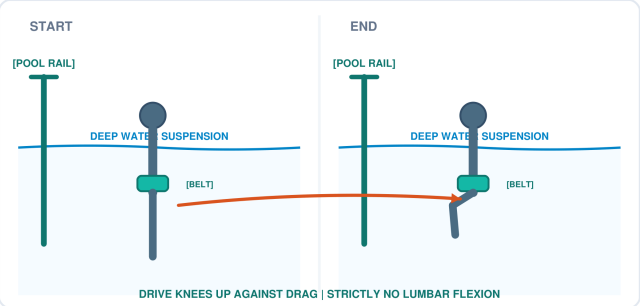
Step 5: Deep Water Core Suspensions

Dosage: 10 mins (4 rounds of 45s work / 45s rest)

Instructional Cues:

- Flotation belt fastened
- Deep-water bicycle sprints
- Alternating knee tucks

Source: Flotation belt on, deep-water bicycle sprints and alternating knee-to-chest tucks. 4 rounds of 45s work / 45s rest.



SAFETY MANDATE: Maintain vertical spinal alignment; no lumbar hyperextension.

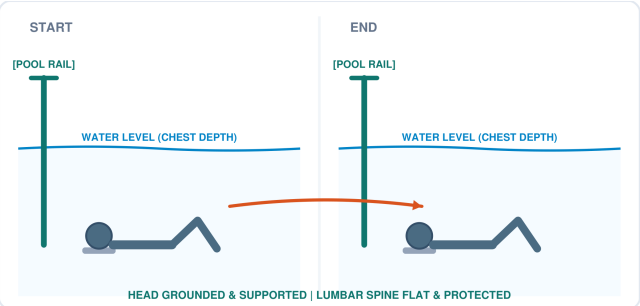
Step 6: Hydro Decompression & Stretches

Dosage: 8 mins

Instructional Cues:

- Calf stretch against pool wall
- Gentle hamstring elongation
- Passive buoyant traction

Source: Calf and hamstring pool-wall stretches, passive buoyant traction.



SAFETY MANDATE: Gentle stretch only; do not bounce or force joint ranges.

H-EXP2: 75-Minute Multi-Planar Hydro Endurance & Kinetic Chain Overhaul

[EXPERT / HIGH RESISTANCE]
Duration: 75 Minutes

Equipment Needed: Heavy Loop Bands | Pool Rail | Water Dumbbells | Pool Noodle

S1: 12m	S2: 15m	S3: 13m	S4: 12m	S5: 13m	S6: 10m

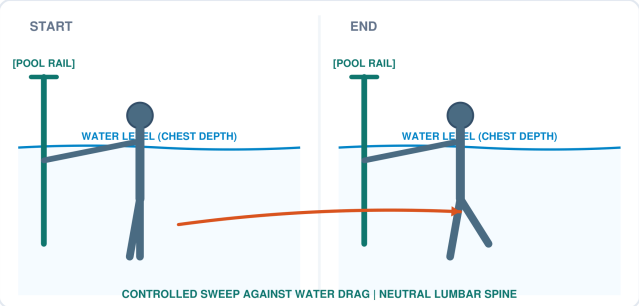
Step 1: Warm-Up & Hydrodynamic Mobilization

Dosage: 12 mins

Instructional Cues:

- Forward, backward, carioca strides
- Hold rail for ankle stability
- Chest-depth water resistance

Source: 400m water stride circuit (forward, backward, carioca/side-step) holding rail for ankle stability.



SAFETY MANDATE: Hold pool rail for right ankle stability; avoid sudden direction changes.

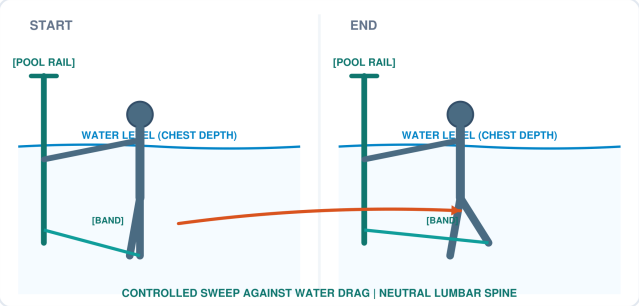
Step 2: Banded Lateral Walk-Outs & Monster Walks

Dosage: 15 mins (4 sets x 15 steps per direction)

Instructional Cues:

- Loop band around ankles
- Chest-deep water
- Step against band tension

Source: Loop band around ankles in chest-deep water. Step against band tension and water resistance. 4 sets x 15 steps per direction.



SAFETY MANDATE: Keep feet pointing straight ahead; prevent knees caving inward.

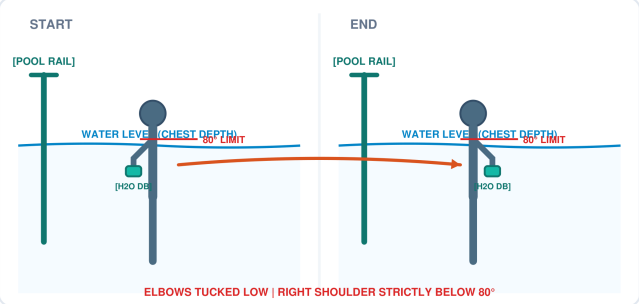
Step 3: Hydro Dumbbell Torso Rotations & Submerged Presses

Dosage: 13 mins (3 sets x 12 reps)

Instructional Cues:

- Dumbbells submerged at chest depth
- Thoracic rotation only
- Cervical spine strictly locked

Source: Hold hydro dumbbells at chest depth. Rotate torso within safe thoracic limits while cervical spine remains strictly locked. 3 sets x 12 reps.



SAFETY MANDATE: Strictly no cervical rotation; neck moves as one unit with sternum.

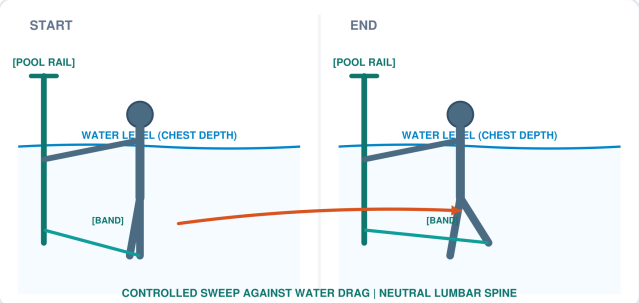
Step 4: Rail-Supported Explosive Glute Kickbacks

Dosage: 12 mins (4 sets x 12 reps per leg)

Instructional Cues:

- Band anchored to rail
- Drive hip into glute extension
- Brace core wall

Source: Band anchored to pool rail. Drive hip back into terminal glute extension without lumbar hyperextension. 4 sets x 12 reps per leg.



SAFETY MANDATE: Zero lumbar hyperextension; movement isolated purely to hip extension.

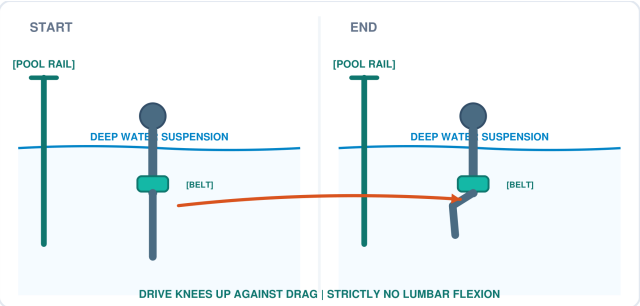
Step 5: Deep Water Interval Treading & Core Stabilization

Dosage: 13 mins (5 intervals: 1-min work / 1-min hold)

Instructional Cues:

- Flotation belt support
- High-cadence treading 1 min
- Recovery hold 1 min

Source: Flotation belt, 1-minute high-cadence treading, 1-minute recovery hold. 5 intervals.



SAFETY MANDATE: Maintain upright posture; stop if extreme fatigue compromises core stability.

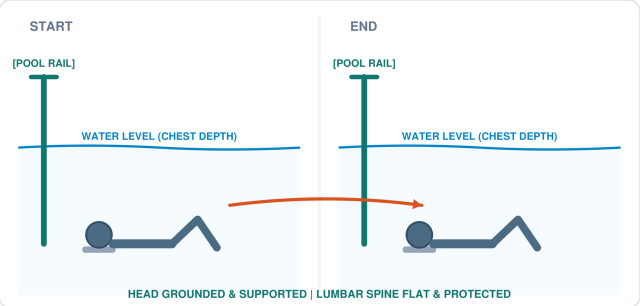
Step 6: Full Hydro Decompression & Active Recovery

Dosage: 10 mins

Instructional Cues:

- Supine flotation elongation
- Gentle rhythmic ankle pumps
- Diaphragmatic breathing

Source: Floating supine elongation, gentle ankle pumps and diaphragmatic breathing.



SAFETY MANDATE: Complete relaxation; ensure head and neck are supported by float.

H-EXP3: 90-Minute Master Hydro Athlete Rehabilitation Protocol

[EXPERT / MAXIMUM RESISTANCE]

Duration: 90 Minutes

Equipment Needed: Full Hydro Resistance Kit (Heavy bands, Hydro paddles, Flotation belt, Noodles, Rail)

S1: 15m	S2: 20m	S3: 20m	S4: 18m	S5: 10m	S6: 7m

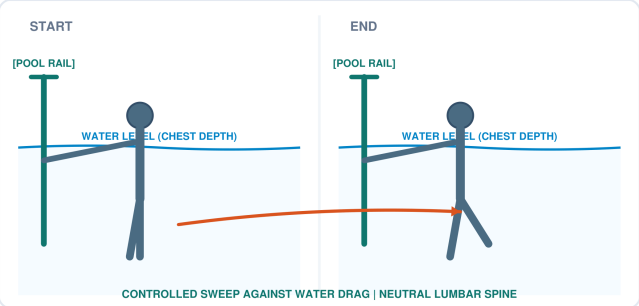
Step 1: Phase 1: Dynamic Hydrodynamic Activation

Dosage: 15 mins

Instructional Cues:

- Progressive multi-speed walking
- Submerged scapular sweeps
- Active ankle dorsiflexion

Source: Progressive multi-speed water walking, gentle submerged scapular sweeps, active ankle dorsiflexion against water drag.



SAFETY MANDATE: Monitor right shoulder range (<80°); support right ankle stability.

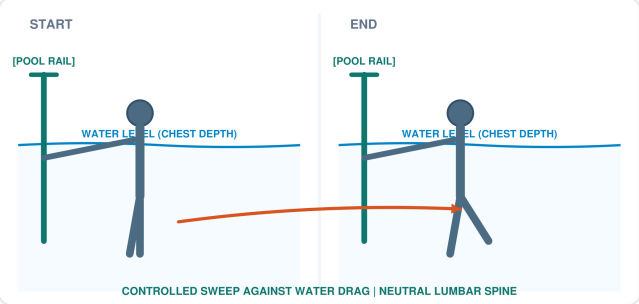
Step 2: Phase 2: Heavy Band Upper-Body Kinetic Chain

Dosage: 20 mins (3 cycles of 10 reps each)

Instructional Cues:

- Rail-anchored banded low rows
- W-rows below shoulder height
- Rotator cuff rotation complexes

Source: Rail-anchored banded low rows, W-rows, internal/external rotation complexes (10 reps each, 3 cycles).



SAFETY MANDATE: Elbows strictly below shoulder line; verify heavy band anchor to pool rail.

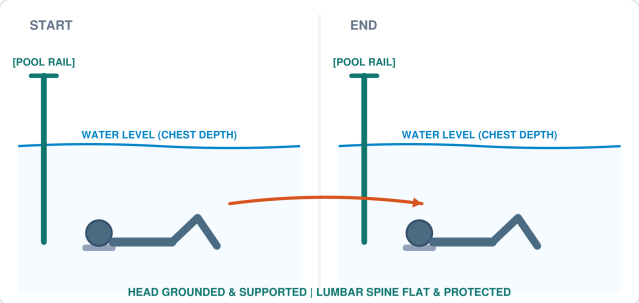
Step 3: Phase 3: Lower Kinetic Chain & Pelvic Power

Dosage: 20 mins

Instructional Cues:

- Banded rail squats (max 45°)
- Submerged knee extensions
- Heavy band side-stepping

Source: Banded rail squats, banded submerged knee extensions, side-stepping against heavy band tension.



SAFETY MANDATE: Max 45° knee flexion; do not allow left knee valgus collapse.

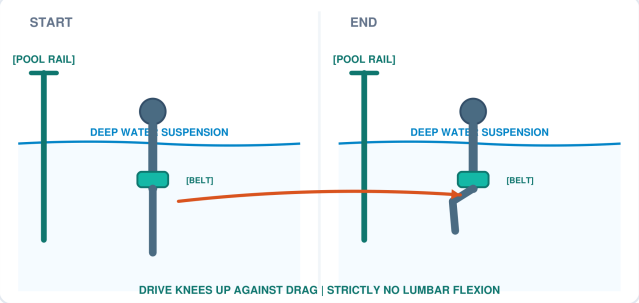
Step 4: Phase 4: Deep-Water Aquatic Core Crucible

Dosage: 18 mins

Instructional Cues:

- Flotation belt suspension
- Knee drives and scissor kicks
- Isometric kickboard downward press

Source: Flotation belt suspended deep-water knee drives, scissor kicks, and isometric kickboard presses under water.



SAFETY MANDATE: Neutral spine locked; no cervical straining or lumbar extension.

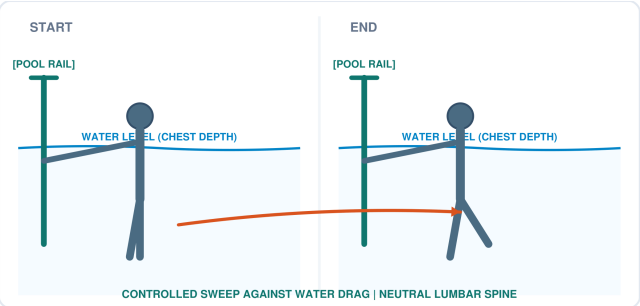
Step 5: Phase 5: Neuromuscular Gait Retraining

Dosage: 10 mins

Instructional Cues:

- High-step water marching
- Focused heel-to-toe push-off
- Contralateral arm swings

Source: High-step walking with focused heel-to-toe push-off and contralateral arm swings.



SAFETY MANDATE: Arms swing naturally below shoulder level; smooth controlled cadence.

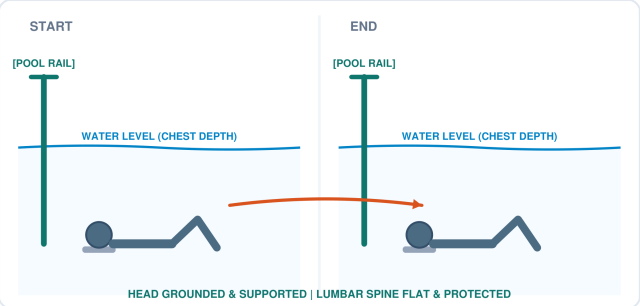
Step 6: Phase 6: Passive Decompression & Thermal Surrender

Dosage: 7 mins

Instructional Cues:

- Noodle supporting cervical spine
- Full buoyant floating
- Complete nervous system downregulation

Source: Cervical-supported noodle float, total muscular downregulation.



SAFETY MANDATE: Full cervical support; allow hydrostatic pressure to unload axial spine.

CLINICAL GOVERNANCE & SAFETY DISCLAIMERS

1. This is appointment preparation and second-opinion context, not a clinical recommendation to act on independently. Discuss with your physiotherapist and neurosurgeon before making changes to your treatment.

2. Any changes to medication or pharmacological therapies must be reviewed, prescribed, and managed by your General Practitioner or managing specialist in consultation with a qualified pharmacist.

3. If you experience sudden severe motor weakness, progressive numbness, fever with joint swelling, or acute changes in bowel or bladder control, seek immediate emergency medical evaluation.